



Virtual competition strategy guide

Julius Kiuru

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Who am I?



- Team captain of the University of Helsinki team at ISC 2025
- PhD student in computational biophysics
- Avid pokémon fan





Before the competition



Before the competition

- Familiarize yourselves with:
 - The applications
 - The tasks
- Plan your workflows beforehand
 - What can be scripted?
- Divide the tasks between yourselves



Example: division of labour

- One person in charge of each task
- Last person:
 - AI task
 - Data analyst
- Make this decision soon



During the competition



Test the competition cluster

- Default thread and process affinities on the cluster?
- Do MPI and OpenMP work together?
- `srun` vs `mpirun`
- Sometimes interactive jobs have less queue time
- If nothing works, try running the applications on CSC's computers



Start working on everything

- Very long tasks
- Get a working model
 - Any results are better than no results
- Communicate actively
 - Shift attention to tasks that need more work



Be creative

- If you were the developer of the application, what would interest you?
- If the instructions/manuals say X is the best way to run this program, try something else
- If you have time, play around with compilation sources, compilation flags, etc.



Collaborate

- Share ideas and approaches
 - Before the competition
 - During the competition?
 - After the competition



Have fun!

- Take advantage of this opportunity
- Don't take things too seriously, enjoy the process
- Make sure to rest
- Don't let the competition take over your life



After the competition



After the competition

- Stay in touch with each other and CSC
- Stay curious and keep learning and developing
- You are brilliant minds!



Summary

- Prepare well before the competition starts
- Start simple and build up
- Be creative
- Have fun
- Good luck!

